

Troubleshooting Rings in an Axial Image

1 Personnel Requirements

No personnel requirements have been defined for this procedure.

2 Procedure Overview

This procedure addresses:

- Rings in an Axial Image
- Image Quality Characteristics & Testing Procedures

3 Preliminary Requirements

3.1 Tools and Test Equipment

There are no special tools identified for this procedure.

3.2 Consumables

There are no identified consumables for this procedure.

3.3 Replacement Parts

There are no identified replacement parts for this procedure.

3.4 Safety

There are no identified general safety statements for this procedure.

3.5 Required Conditions

There are no conditions identified as required for this procedure.

4 Procedure

4.1 Rings in an Axial Image

Using the Example shown in [Figure 1](#), there is a obvious ring in the first image within the group of 4 images. During an Axial series, it is important to know certain parameters about the Scan Prescription. See [Table 1](#)

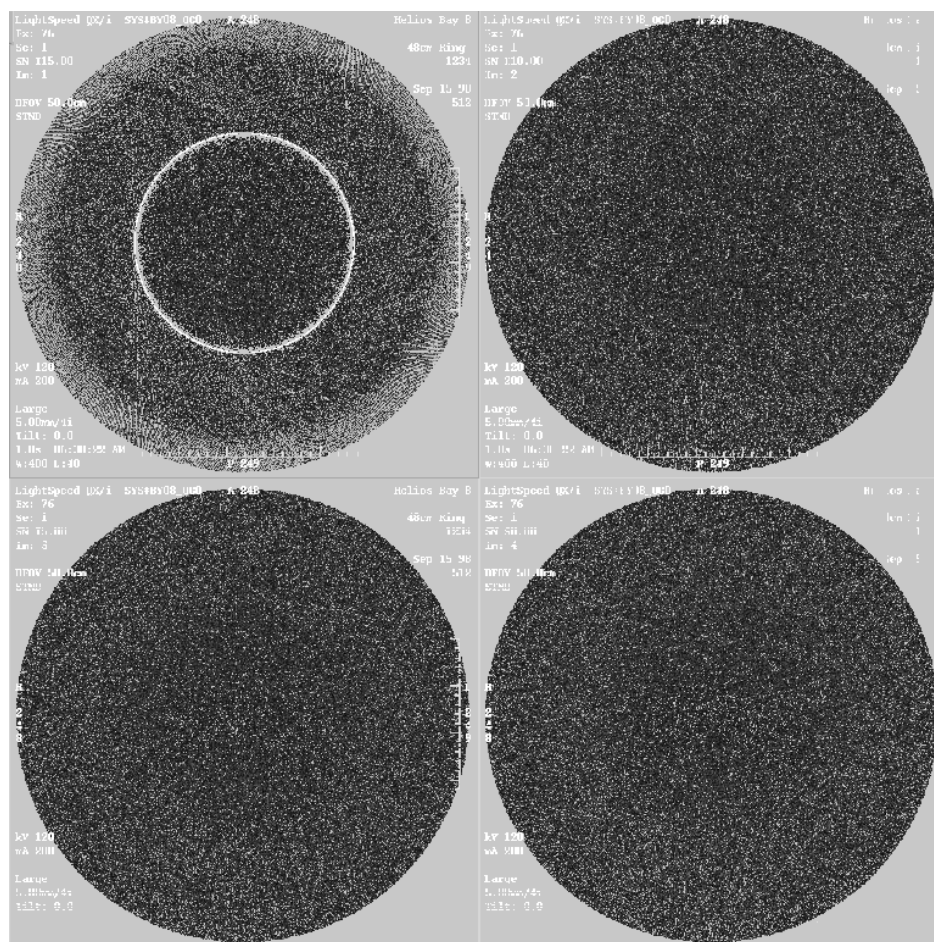
SCAN (Rx) PRESCRIPTION		COMMENTS
Scan Type	Axial	In a Axial mode, there are several options relative to scanning and displaying images, either 4 x i, or 2 x l (where i = thickness) In a 4 x i mode, images are created from the rows used as indicated by slice thickness. <i>This mode should always be used to troubleshoot Detector / DAS related artifacts.</i> In a 2 x i mode Recon actually combines 2 rows of data to produce a thicker slice.
Scan Type	Helical	Images are created using data from all Rows used during data acquisition. (The actual rows being dependent on slice thickness) The amount of data used to reconstruct a image from a row, or row weighting is dependent on table speed.
Head First	Head First	Image 1 consists of the Data in Row 4A, Image 2 consists of the Data in Row 3A Image 3 consists of the Data in Row 2A Image 4 consists of the Data in Row 1A Image 5 consists of the Data in Row 1B, Image 6 consists of the Data in Row 2B Image 7 consists of the Data in Row 3B Image 8 consists of the Data in Row 4B
Feet First	Feet First	Image 1 consists of the Data in Row 4B, Image 2 consists of the Data in Row 3B Image 3 consists of the Data in Row 2B Image 4 consists of the Data in Row 1B Image 1 consists of the Data in Row 1A, Image 2 consists of the Data in Row 2A Image 3 consists of the Data in Row 3A Image 4 consists of the Data in Row 4A
Slice Thickness	4 x 1.25	Uses 4 Center rows of the Detector (D2, D1, D1, D2)
	4 x 2.50	Uses 8 Center rows of the Detector (D4+D3, D2+D1, D1+D2, D3+D4)
	4 x 3.75	Uses 12 Center rows of the Detector (D6+D5+D4, D3+D2+D1,)
	4 x 5.00	Uses 16 rows of the Detector (D8+D7+D6+D5, D4+D3+D2+D1,...)
	8 x 1.25	Uses 16 rows of the Detector (D8, D7, D6, D5, D4, D3, D2, D1, D1, D2...)
	8 x 2.50	Uses 16 rows of the Detector (D8+D7, D6+D5, D4+D3, D2+D1, D1+D2...)
	1 x 1.25	Uses 1 Center row of the Detector (A side D1)
	Thin Twin	Uses 2 Center rows of the Detector (D1, D1)

Table 1: Troubleshooting a Ring in an Axial Image

SCAN (Rx) PRESCRIPTION		COMMENTS
Technique	KV vs. Slice Thickness	The DAS Converter boards have 31 pre-amp gains that are selected by selected technique. These gains are not User selectable, however it is important to know if a Artifact only appears at the same pre-amp gain. The pre-amp gain used can be found within the scan header by using Scan Analysis.
Focal Spot Size	Small vs. Large	Technique dependent small spot $\leq 24KW$, Large spot $> 24KW$
Calibration Used	Small vs. Large	Needed to know if artifact may be caused by cal vectors

Table 1: Troubleshooting a Ring in an Axial Image

Figure 1: Example of Bad Channel in Axial Image Series

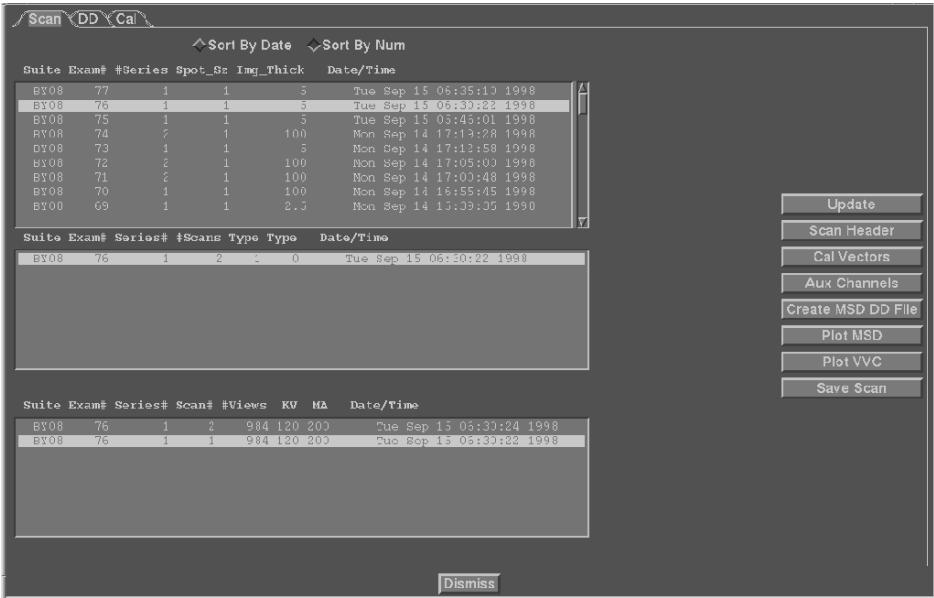


Looking at the Example in [Figure 1](#), just by knowing the scan series was using Axial 4 x 5.00 Mode and was done Head first (Annotation of image vs. Table position), it can be determined that the Ring is probably in Row 2A of the Scanfile.

To confirm this, use Scan Analysis and plot the Means & Standard Deviations (**MSD**) of the selected Exam, Series, and Scan Number. From the Service Desktop:

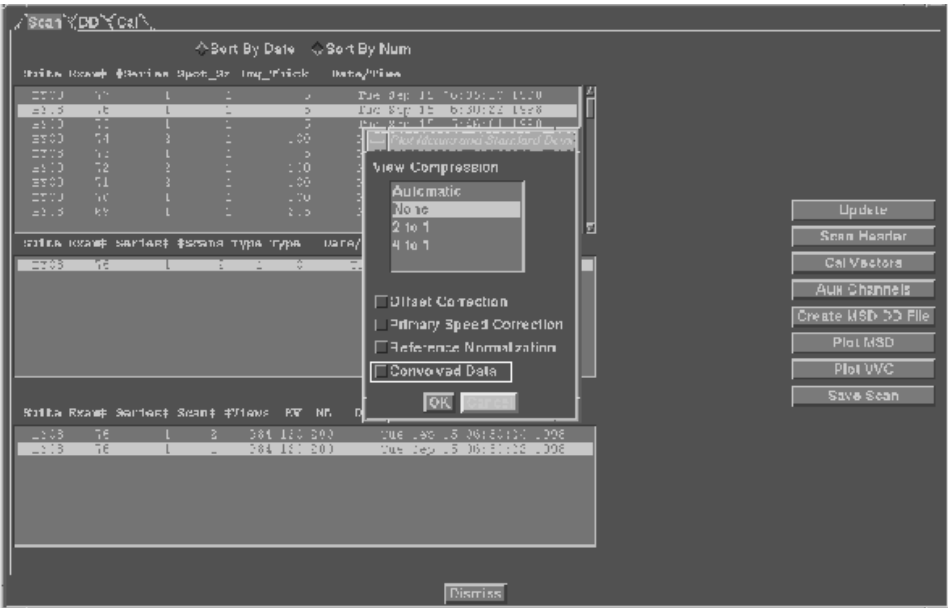
- 1.) Select UTILITIES.
- 2.) Select TOOLS.
- 3.) Select SCAN ANALYSIS.

Figure 2: Scan Analysis GUI

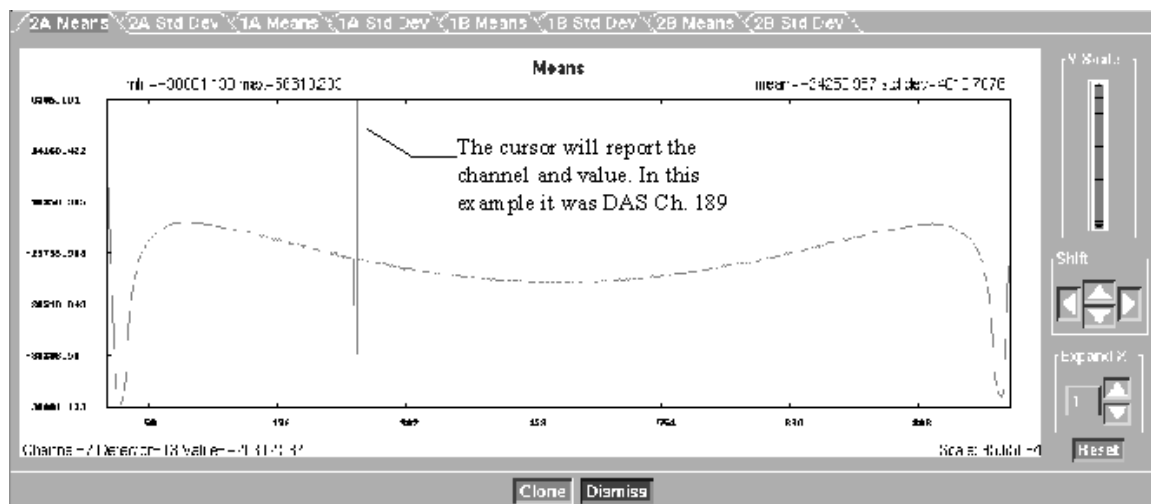


Select an EXAM NUMBER, SERIES, and SCAN NUMBER. Remember that 1 scan, in a 4 x i mode, is made up of 4 Data sets (or rows), which produces 4 images. In the example, images 1 through 4 are created from scan 1. Once the scan is highlighted, select PLOT MSD.

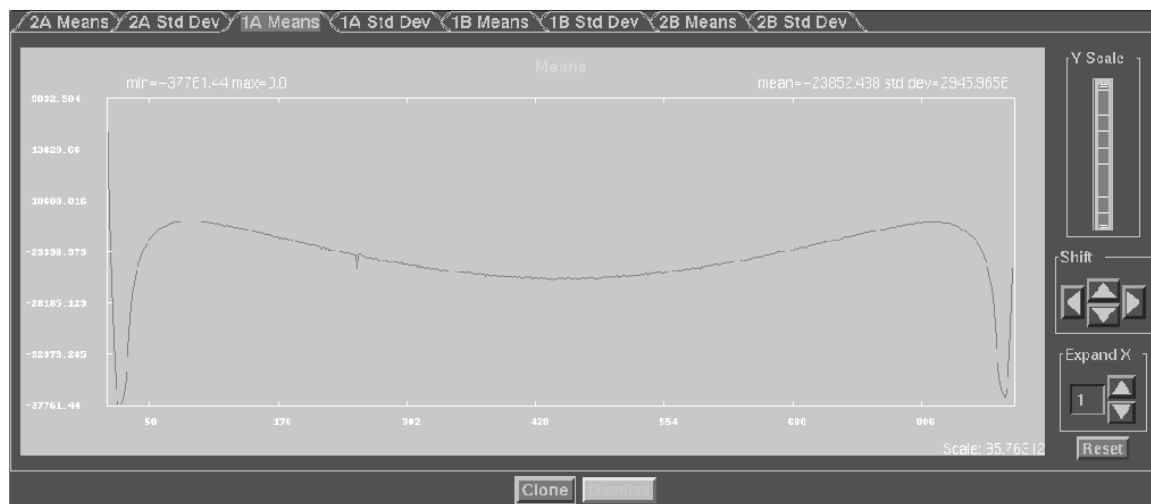
Figure 3: Plot MSD GUI



Leave the view compression defaulted to NONE, but choose CONVOLVED DATA, which will identify a ring type artifact better in the resultant plot. Select OK.

Figure 4: Scan Analysis (2A) Means & Standard Deviation GUI

The Scan Analysis Tool will first plot Row 2A. Any of the 4 rows of Means and Standard Deviations can be viewed by selecting the appropriate tabs (see [Figure 4](#)). Select the tab that indicates the row where the ring is expected based on your initial observations. It may be necessary to adjust the level to find a spike in the data or view other rows. Look for any abnormal spikes.

Figure 5: Scan Analysis (1A) Means & St. Dev. GUI

In Row 2A (see [Figure 4](#)), the ring is apparent. Notice the large spike in the data on channel 189. Row 1A (see [Figure 5](#)) has a small spike on channel 189 that is a result of capacitive discharge from row 2b channel 189. The small spike can be ignored. It is a product of the major spike on row 2B. Rows 1B and 2B look good. See [Figures 6 and 7](#).

Figure 6: Scan Analysis (1B) Means & St. Dev. GUI

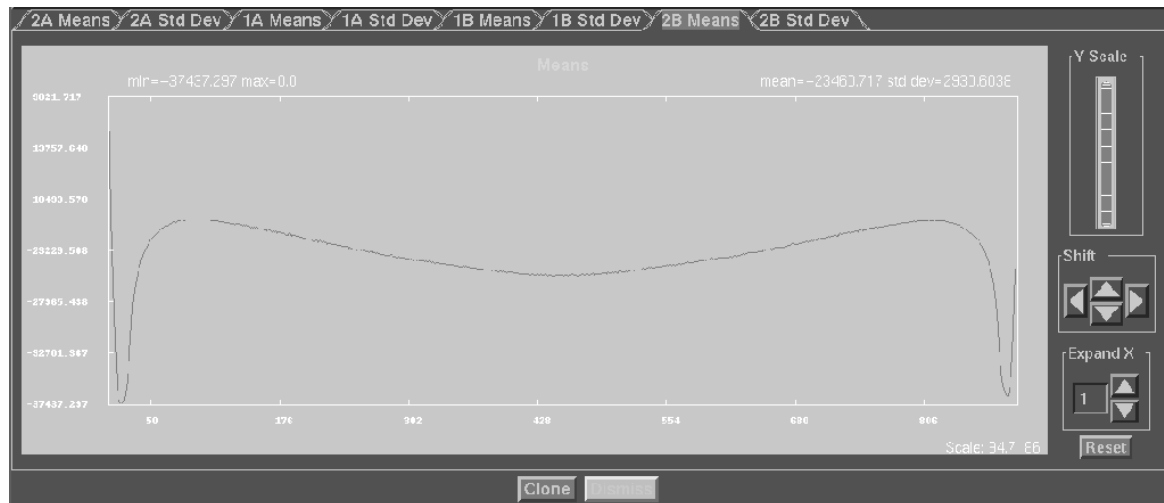


Figure 7: Scan Analysis (2A) Means & St. Dev. GUI

Now the Ring has been verified. It is in Row 2B and is on **DAS** Channel 189.

From within Scan Analysis, Cal vectors can be plotted to see if the bad **DAS** channel is present.

Determine if the ring is caused by a particular acquisition mode by scanning a phantom using different modes or slice thickness'. The example was scanned in a 4 x 5.00 mode and the ring appeared on Row 2A, therefore, the Detector rows or diodes used were D8+D7+D6+D5. They produce row 2A. If another scan was taken at 4 x 1.25 mode, then the first image, or row 2A data would be acquired from Detector row D2. If the ring is a hard failure (consistent every time) and if after changing slice thickness the ring does away, the ring may have been caused by a suspect detector. Remember to use 8 x 1.25 mm is the 8 slice system configuration. *Perform further detector verification before replacing a detector.* If the ring is still present, the problem could still be the detector, but may be a **DAS** board or elastomer interface connection.

By using the DAS / DETECTOR Architecture Tool, found in the pull-down menu under FILE, select the tool. A TTY window appears and prompts for a Detector Row and **DAS** Channel. The program will display the associated **DAS** Converter board, Detector channel, module number, elastomer number, and other important information. Find the associated **DAS** converter board number. If you have multiple rings, look for patterns (Converter Bd, Row, etc.).

If **DAS** is Suspect, swap filter Cards and repeat Scan & Analyze. When swapping cards make sure you swap with a card 4 slots away, slot 12 with slot 16 for example. This ensures the data from FET multiplexing is not on the same card.

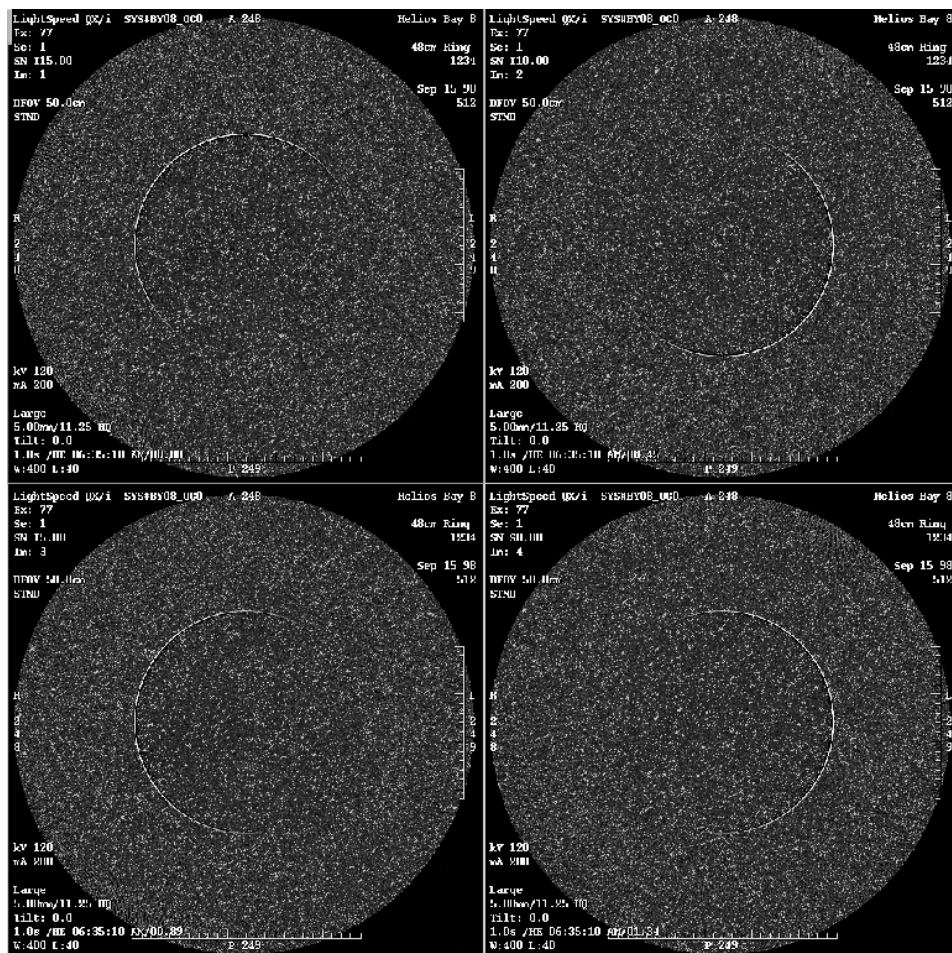
- 1.) If bad channel follows bd. replace the bd.
- 2.) If bad channel stays at same location, problem area could be:
 - a.) **DAS** Backplane
 - b.) **DAS** Backplane to Elastomeric connection point
 - c.) Elastomeric to Detector Flex connection point
 - 1.) Bad Detector (Replace)
 - 2.) Bad Flex (part of detector, replace Detector)
 - 3.) Bad Flex to Detector connection point (part of detector, replace Detector)
 - 4.) Bad Detector Channel (part of detector, replace Detector)
 - 5.) Bad Detector Cell (part of detector, replace Detector)

Troubleshoot Ring Artifacts in a Helical series by duplicating the problem in a Axial Mode. If the Helical data is used to troubleshoot, the bad data can be in any row since all 8 rows of data can be

used to produce a Helical image and the number of views to produce a helical image are weighted differently based on table speed and pitch.

Note that the Helical images are a result of the same bad **DAS** Channel as the Above Axial example. The hard ring in the Axial example appears as partial arcs in the Helical images.

Figure 8: Example of Bad Channel in Helical Image Series



4.2 Image Quality Characteristics & Testing Procedures

4.2.1 What to Check for IQ

LightSpeed 16 slice systems use 16 rows of detectors with up to 16 rows of output, which produces up to sixteen images at one rotation. LightSpeed 16 slice systems also utilize data from all selected rows in helical reconstruction by using different weighting schemes to produce better quality helical images. LightSpeed 16 slice systems are also designed to perform longer helical scans at lower mAs. Because of the differences, when checking the image quality (IQ), we need to pay attention to those artifacts/appearance that are unique to the LightSpeed 16 slice scanner. The following are some of the examples:

- 1.) Ring/band artifacts may appear as partial or non-uniform ring/band in helical images. When this happens, axial scans may need to be taken to localize the bad **DAS** channel(s) or detectors.
- 2.) If artifacts happen in images of one of the outer rows, it may indicate a misalignment of beam in Z-direction (**BOW**). Again, axial scans may need to be taken to further isolate the problems if the problems happened in the helical images.
- 3.) LightSpeed uses 16 detectors that cover more space in Z-direction than CT/i. When using thicker slice configurations (2.5mm/8i), axial images made by the two outer rows are more susceptible to cone beam artifact (an example of a cone beam artifact is shading off a rib into the liver). Also, the use of peristaltic correction can increase the cone beam effect. In general, the use of thinner slice configurations reduces cone beam artifact. Also, helical mode will help reduce cone beam artifact, with HQ mode having less cone beam artifact than HS mode.

Specifically, the IQ checks may include the following:

- 1.) Alignment
 - a.) **MTF** with GE Performance phantom, both axial and helical images, comparing with spec.
 - b.) Aliasing test, both cold and hot.
 - c.) Z-Beam motion.
- 2.) Noise/Artifacts
 - a.) Noise measurement with 20 cm water phantom, comparing with spec.
 - b.) Cone beam artifact.
- 3.) Microphonics - Check image quality (lack of rings, streak and/or center artifacts).
- 4.) Clever **DAS** Gain - Confirm that the **DAS** gains used in the patient scans match those in the cals.
- 5.) **CT** Number Uniformity - Check both water and poly phantoms' **CT** number uniformity, comparing with spec.

4.2.2 How to Check Image Quality

For artifact definitions and numerical measurements, see Installation Manual. For Image Troubleshooting, see Installation Manual. However, if artifacts happened in helical images, take axial scans to further isolate the problems.

ALIGNMENT

- **MTF** - Values should be similar to those of CT/i.
 - Phantom: GE performance phantom (if available). The wire section of the GE performance phantom is not thick enough to have 2 10mm thick images. Two scans need to be done to verify both sides (2A1A and 1B2B) of the detector.
 - Data Collection:
 - 1.) Center the phantom in x-y direction.

- 2.) Landmark (in z-direction) at the center of the wire.
- 3.) Take scans per [Table 2](#).

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
small	120/440	10mm/2i	Ax/0.8sec	25cm/std. 10cm/edge	S0 - S10	2	0	@S0
small	120/440	10mm/2i	Ax/0.8sec	25cm/std. 10cm/edge	I10 - S0	2	0	@S0
small	120/130	10mm/2i	Ax/2.0sec	25cm/std. 10cm/edge	S0 - S10	2	0	@S0
small	120/130	10mm/2i	Ax/2.0sec	25cm/std. 10cm/edge	I10 - S0	2	0	@S0
small	120/440	5mm/ 7.5mm/ sec	HQ 1sec	25cm/std. 10cm/edge	I5 - S5	4	0	@S0

Table 2: Alignment Scan Settings

- Data Analysis: Use the "Image Resolution" button in the "Image Analysis" tool kit to measure system **MTF**, compare with spec. Use images at S0.
- **ALIASING** - Check ISO alignment, and check if hot ISO was executed correctly.
 - Phantom: GE QA phantom, section No. 1.
 - Data Collection:
 - 1.) Fastcal 120 kV, head bowtie.
 - 2.) Let tube cool 30 minutes after last fastcal or scanning.
 - 3.) Center the GE QA phantom and take scans per [Table 3](#).
 - 4.) Take heating scans using the protocol for QOEC tube heating.
 - 5.) Take scans in (4) again.

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
small	120/200	2.5mm/4i	Ax/1sec	12.8cm/bone	I3.8 - S3.7	1	0	all
small	120/210	2.5mm/4i	Ax/1sec	12.8cm/bone	I3.8 - S3.7	1	0	all

Table 3: Aliasing Scan Settings

- Data Analysis: Inspect images visually. Both cold and hot tube images should be lack of aliasing artifacts.
- **Z-BEAM MOTION** - If artifacts happened in one of the outer rows, it may suggest that the **BOW** is not aligned correctly.
 - Phantom: GE QA Phantom
 - Data Collection:
 - 1.) Fastcal 120kV, head Bowtie
 - 2.) Let tube cool 1 hour after last fastcal or scanning

3.) Center the QA phantom water section, take scans per [Table 4](#).

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
small	120/200	5mm/4l	Ax/1sec	25cm/std.	I7.5 - S7.5	1	0	all
small	120/100	5mm/4l	Ax/4sec	25cm/std.	I7.5 - S7.5	1	0	all
small	120/200	1.25mm/4l	Ax/1sec	25cm/std.	I1.9 - S1.8	1	0	all
small	120/100	1.25mm/4l	Ax/4sec	25cm/std.	I1.9 - S1.8	1	0	all

Table 4: Beam Motion Scan Settings

4.) Scan a series of 30 scans at 4sec with 4sec ISD per [Table 5](#).

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
small	120/200	5mm/4i	Ax/4sec	25cm/std.	I7.5 - S7.5	30	0	None

Table 5: Scan Settings (4sec ISD)

5.) Repeat scans in (4) again.

Data Analysis: Check images visually, pay attention to the two outer rows. No failing ring and/or band artifacts should occur

NOISE

Image noise should meet the spec.

- Phantom: GE QA phantom.
- Data Collection:
 - 1.) Fastcal, 120kV, Head Bowtie.
 - 2.) Center the QA phantom water section, and take scans per [Table 6](#).

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
small	120/260	10mm/2i	Ax/1sec	25cm/std.	I5 - S5	1	0	all
small	120/190	10mm/ 15mm/sec	HE-HQ 1sec	25cm/std.	I10 - S10	2	0	@S0

Table 6: Noise Scan Settings

Data Analysis: Place **ROI** in the center and measure image noise (standard deviation). The spec is $2.7 < \text{Stdv} < 3.3$ for both axial and helical scans at the above techniques with peristaltic off.

CONE BEAM ARTIFACT

An example of a cone beam artifact would be shading off a rib into the liver. The greater the cone angle (slice thickness in Z-direction), the more severe the cone beam artifact. Cone beam artifact is typically more pronounced in the images from the two outer rows, where the **POR** is not as accurate as in the inner rows, especially when using thicker slices. Images obtained using thinner slice configurations should improve image quality in terms of cone beam artifact. Helical reconstruction uses all selected row outputs to minimize cone beam artifacts.

CLEVER DAS GAIN

A multiple **DAS** gain scheme is used to reduce the low signal image noise. For body bowtie, scan techniques (kV/Collimation/Spot Size) are further divided into different mA ranges based on the detector collimation. For example, for the technique of 120kV/bbt/4x500/LS we have three different mA ranges:

- 210 - 240 mA (gain 17)
- 250 - 300 mA (gain 21)
- 310 - 440 mA (gain 31).

Artifacts (ring, band and center spot) may be caused by a failure of a **DAS** channel at one of the gains but not all of them (i.e., an image artifact at a certain mA range but not others). Run **DAS** check to identify or exclude **DAS** problems.

140KV/380MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	380 (250 - 380)	31	1.19 (lockout @320mA)
4x500	240 (180 - 240)	24	0.97
4x375	380 (250 - 380)	29	0.95
4x375	240 (180 - 240)	18	0.96
4x250	380 (250 - 380)	19	0.96
4x250	240 (180 - 240)	13	0.92
4x125	380 (180 - 380)	10	0.94

Table 7: DAS Gains, Body Filter, Large Spot (140kV/380mA)

120KV/440MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	440 (310 - 440)	31	0.96
4x500	300 (250 - 300)	21	0.96
4x500	240 (210 - 240)	17	0.94
4x375	440 (310 - 440)	24	0.94
4x375	300 (250 - 300)	16	0.95
4x375	240 (210 - 240)	13	0.97
4x250	440 (310 - 440)	16	0.92
4x250	300 (210 - 300)	11	0.94
4x125	440 (280 - 440)	8	0.96
4x125	270 (210 - 270)	5	0.95

Table 8: DAS Gains, Body Filter, Large Spot (120kV/440mA)

100KV/420MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	420 (250 - 420)	19	0.95
4x375	420 (250 - 420)	15	0.94

Table 9: DAS Gains, Body Filter, Large Spot (100kV/420mA)

100KV/420MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x250	420 (250 - 420)	10	0.94
4x125	420 (250 - 420)	5	0.93

Table 9: DAS Gains, Body Filter, Large Spot (100kV/420mA) (Continued)

80KV/400MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	400 (310 - 400)	9	0.98
4x375	400 (310 - 400)	7	0.96
4x250	400 (310 - 400)	5	0.90
4x125	400 (310 - 400)	3	0.73

Table 10: DAS Gains, Body Filter, Large Spot (80kV/400mA)

140KV/170MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	170 (130 - 170)	17	0.95
4x500	120 (10 - 120)	13	0.91
4x375	170 (120 - 170)	13	0.97
4x375	110 (10 - 110)	9	0.91
4x250	170 (120 - 170)	9	0.94
4x250	110 (10 - 110)	6	0.91
4x125	170 (10 - 170)	5	0.85

Table 11: DAS Gains, Body Filter, Small Spot (140kV/170mA)

120KV/200MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	200 (150 - 200)	14	0.98
4x500	140 (90 - 140)	10	0.96
4x500	80 (10 - 80)	6	0.92
4x375	200 (170 - 200)	11	0.96
4x375	160 (100 - 160)	9	0.94
4x375	90 (10 - 90)	5	0.96
4x250	200 (170 - 200)	8	0.87
4x250	160 (10 - 160)	6	0.93
4x125	200 (10 - 200)	4	0.88

Table 12: DAS Gains, Body Filter, Small Spot (120kV/200mA)

100KV/240MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	240 (10 - 240)	11	0.92
4x375	240 (10 - 240)	8	0.96
4x250	240 (10 - 240)	6	0.86
4x125	240 (10 - 240)	3	0.85

Table 13: DAS Gains, Body Filter, Small Spot (100kV/240mA)

80KV/300MA	MA BREAK POINT (MA RANGE)	DAS GAIN	BOWTIE OVR FACTOR AR ISO
4x500	300 (240 - 300)	7	0.87
4x500	230 (10 - 230)	5	0.94
4x375	300 (10 - 300)	5	0.90
4x250	300 (10 - 300)	4	0.76
4x125	300 (10 - 300)	3	0.48

Table 14: DAS Gains, Body Filter, Small Spot (80kV/300mA)

MICROPHONICS

Scan phantom at low signal level to test insensitivity of the **DAS**, Detector, and interconnect system to noise generated by vibration sources. Sources of vibration could be gantry rotation, tube rotor, fans pumps, dust in DAS etc.

- Phantom: 48cm Poly phantom
- Data collection:
 - Fastcal, 120 kV, Body Bowtie
 - Center 48cm poly phantom, and take scans per [Table 15](#).

Cal	kV/mA	Slice thickness	Scan mode/time	Recon FOV / alg	Scan plane	# of scans	Tilt	Images analyzed
large	120/200	5mm/4i	Ax/1sec	48cm/std	11.9 - S1.8	30	0	all

Table 15: Microphonics Scan Settings

- Data Analysis: Images should have no visible rings/bands, streak and center artifacts

CT NUMBER UNIFORMITY

- Phantom: 5 inch water, GE QA, 35cm Poly and 48 Poly phantoms.
- Data Collection: Perform 1X series scans.
- Data Analysis: Use the "Series Means" button in the "Image Analysis" tool kit to analyze the CT number uniformity.

5 Finalization

No finalization is required.